



Submittal Review Response

Project Name: Hilo WWTP Rehabilitation and Replacement Project Phase 1
Submittal No.: 03200-003.0
Date: 9/25/2025

Client: County of Hawai'i **Carollo Project No.:** 203975
Contractor: Nan, Inc.
Submittal Name: Concrete Reinforcing Screening Wall and Top Slab
Reviewed By: Marissa Kurniawan, Hipom Caleb Che

SUBMITTAL REVIEW

Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.

Approved	☐	No Exceptions
	☐	Make Corrections Noted - See Comments
	☐	Make Corrections Noted - Confirm
Not Approved	☒	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

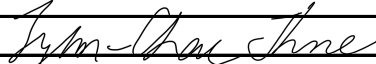
- Comments are provided in the attachment. Verify and address all comments.
- Contractor is responsible for providing waterstops at locations indicated in contract drawings.
- For submittal page 1, coordinate and provide reinforcement at walls and channels for pipe openings/penetrations per P304/TYP, S180/TYP, and mechanical drawings.
- Submittal page 1 does not show wall-to-bottom slab joint profiles around drain pipes. At interfaces between wall-to-bottom slab around drain pipes on contract drawing 02-S-01-103 (indicated by key note 17 and 18), provide reinforcement per detail 60/02-S-01-512. Additionally, at corresponding locations on the bottom slab, confirm that reinforcement per S158/TYP is provided.
- The reinforcement detailing between slab openings on the screening top slab plan view is unclear (clouded areas on submittal page 3 and 4). Revise to clearly show 3 #6 bars top & bottom total for each clouded area on plan shop drawings per contract drawing 02-S-01-104, keynote 9 and detail 10/02-S-01-501.
- Screening top slab reinforcement plan view drawings show horizontal reinforcement spanning long way and short way going through small openings. Verify and coordinate reinforcement arrangement in areas with these openings and revise shop drawings as necessary.
- Contractor shall coordinate the final opening dimensions with the bar screen equipment dimensions. Reinforcement dimensions in corresponding areas shall be revised accordingly with the final opening dimensions.

☒ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. Project No.: WW-4705R
Project Name: Hilo WWTP Phase 1 Submittal Number: 03200 - 003.0
Submittal Title: CONCRETE REINFORCING SHOP DRAWINGS-S ☐ For Information Only
TO: County
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec: 03200	Paragraph: 1.04
Authorized By: CMC Rebar	Date Submitted: 09/17/2025

Submittal Certification	
Check Either (A) or (B):	
☒ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: 	
Printed Name and Title: Jyun-Cheng Jhuo, Project Engineer	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM:
Date Received PM/CM to Reviewer:
Date Received Reviewer to PM/CM:
Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/17/2025
SPECIFICATION SECTION # 03200
SPECIFICATION Concrete Reinforcing
PARAGRAPH 1.04
DRAWING Shop Drawing
SUBCONTRACTOR n/a
SUPPLIER CMC Rebar
MANUFACTURER n/a

CERTIFIED BY CQCM or Designee : 

SECTION 03200
CONCRETE REINFORCING

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes:
 - 1. Reinforcing bars:
 - a. Carbon steel.
 - 2. Thread bars.
 - 3. Bar supports.
 - 4. Tie wires.
 - 5. Welded wire fabric.
 - 6. Mechanical reinforcing bar couplers.

1.02 REFERENCES

- A. American Concrete Institute (ACI):
 - 1. 318 - Building Code Requirements for Structural Concrete and Commentary.
 - 2. SP-66 - ACI Detailing Manual.
- B. American Iron and Steel Institute (AISI).
- C. American Welding Society (AWS):
 - 1. D1.4 - Structural Welding Code - Reinforcing Steel.
- D. ASTM International (ASTM):
 - 1. A493 - Standard Specification for Stainless Steel Wire and Wire Rods for Cold Heading and Cold Forging.
 - 2. A615 - Standard Specification for Deformed and Plain Carbon Steel Bars for Concrete Reinforcement.
 - 3. A706 - Standard Specification for Low-Alloy Steel Deformed and Plain Bars for Concrete Reinforcement.
 - 4. A1064 - Standard Specification of Carbon-Steel wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete.
- E. Concrete Reinforcing Steel Institute (CRSI):
 - 1. Manual of Standard Practice.
- F. ICC Evaluation Service (ICC-ES):
 - 1. AC133 - Acceptance Criteria for Mechanical Connector Systems for Steel Reinforcing Bars.

1.03 DEFINITIONS

- A. Architectural Concrete: Concrete surfaces that will be exposed to view in the finished work.
 - 1. Additionally, for purposes of this Section, includes:
 - a. Concrete surfaces that are designated to receive paints or coatings.
 - b. Exposed concrete in open basins, channels, and similar liquid containing structures: Surfaces shall be considered exposed to view if located above a line 2 feet below the normal operating water surface elevation in that structure.
- B. Bars: Reinforcement or reinforcing bars as specified in this Section.
- C. Evaluation Report: Report prepared by ICC-ES , or by other testing agency acceptable to the Engineer and to the Building Official, that documents testing and review of a product to confirm that it complies with the requirements of designated ICC-ES Acceptance Criteria, and its acceptance for use under the Building Code specified in Section 01410 - Regulatory Requirements.
- D. Give Away Bars: Reinforcing bars that are not required by the Contract Documents, but are installed by the Contractor to provide support for the required reinforcing bars.
- E. Wire Supports: Metal reinforcing supports constructed of steel wire as specified. Includes individual high chairs, continuous high chairs, bolsters and other similar configurations and shapes.

1.04 SUBMITTALS

- A. General:
 - ✓ 1. Submit in accordance with Section 01330 - Submittal Procedures.
 - 2. Changes to reinforcement in Contract Documents:
 - a. Indicate in a separate letter submitted with shop drawings any changes to reinforcement indicated on the Drawings or specified.
 - b. Such changes will not be acceptable unless Engineer has accepted them in writing.
- B. Product data:
 - ✓ 1. Bar supports:
 - a. Wire bar supports:
 - 1) Schedule of support materials to be provided and locations of use.
 - b. Precast concrete bar supports ("dobies"):
 - 1) Manufacturer's data indicating compression strength of concrete and confirming dimensions and thickness(es).height(s) to be provided for each location where used.
 - 2. Mechanical reinforcing bar couplers. For each type and/or series to be provided:
 - a. Evaluation Report documenting compliance with the requirements of ICC-ES AC133.

- b. Details, properties, and dimensions of couplers. Include type or size identification, and bar size(s) and grade(s) for which the coupler is suitable.
- c. Manufacturer's installation and testing instructions.
- d. Manufacturer's statement that products installed in accordance with manufacturer's recommended procedures will develop strengths and limit slip as specified in this Section.

 C Shop drawings:

- 1. Reinforcement shop drawings:
 - a. Submit drawings showing bending and placement of reinforcement required by the Contract Documents.
 - b. Clearly indicate structures or portions of structures covered by each submittal.
 - 1) Submit reinforcement shop drawings for each structure as a complete package. Submittals addressing only a portion of a structure will be rejected and returned without review, unless such presentation is accepted by Engineer in advance.
 - c. Shop drawings shall conform to the recommendations of the CRSI Manual of Standard Practice and ACI SP-66.
 - d. Use the same bar identification marks on bending detail drawings, placement drawings, and shipping tags.
 - e. Submittals consisting solely of reinforcing bar schedules, without accompanying placement drawings, will not be accepted unless accepted under prior written agreement with Engineer.
- 2. Reinforcement placement drawings:
 - a. Clearly show placement of each bar listed in the bill of materials, including additional reinforcement at corners and openings, and other reinforcement required by details in the Contract Documents.
 - b. Clearly identify locations of reinforcement with coatings (e.g., galvanized or epoxy) and with yield strength other than ASTM A615, Grade 60.
 - c. Show anchor bolt locations based on anchor bolt templates for approved equipment.
 - d. Show splice locations.
 - e. Show locations of mechanical reinforcing couplers, if used.
- 3. Reinforcement fabrication drawings:
 - a. If bend types or nomenclature differs from that recommended in the CRSI Manual of Standard Practice, provide details showing bend types and dimensional designations.
Clearly identify reinforcement with coatings and with yield strength other than ASTM A615, Grade 60.

D. Samples (when requested by Engineer):

- 1. Bar supports/wire reinforcement supports: Samples of each type of chair and bolster proposed for use. Submit with letter stating where each type will be used.
- 2. Precast concrete bar supports: Samples of each type of precast support proposed for use. Submit with letter stating where each will be used.

- E. Test reports:
 - 1. Certified copy of mill test for each steel used. Show physical properties and chemical analysis.
 - a. Mill test reports may be submitted as record documents at the time the reinforcement from that heat of steel is shipped to the site.
 - b. In such cases, submit certificates under the shop drawing submittal number with the letter "R" (for record date) appended to the end (e.g., of the reinforcement was submitted as 03200-002-1, deliver the associated mill certificate as submittal 03200-002-1R).
 - 2. Mechanical reinforcing bar couplers:
 - a. Current Evaluation Report confirming that couplers provide specified tension and compression strength and conform to specified limits on total slip within the coupler.
 - b. Certified copy of mill tests for heat(s) of steel incorporated into the reinforcing bar couplers shipped.
 - c. For threaded sleeve type couplers, heat treatment lot numbers for each shipment.
- F. Manufacturer's instructions:
 - 1. Mechanical reinforcing bar couplers:
 - a. Manufacturer's installation instructions.
 - b. Manufacturer's instructions for confirmation testing of couplers after reinforcing bars have been inserted into the couplers.
- G. Special procedures:
 - 1. Welding procedures conforming to AWS D1.4 for reinforcement to be field welded.
 - a. Procedures qualification record.
- H. Qualifications statements:
 - 1. Welder qualifications.
- I. Closeout documents:
 - 1. Field quality control and inspection reports.
 - 2. Field quality assurance special inspection and testing reports.

1.05 DELIVERY, STORAGE, AND HANDLING

- A. Packing and shipping:
 - 1. Deliver bars bundled and tagged with identifying tags.
- B. Acceptance at site:
 - 1. Reinforcing bars: Deliver reinforcing bars lacking grade identification marks with letter containing manufacturer's guarantee of grade.

1.06 SEQUENCING AND SCHEDULING

- A. Bar supports:
 - 1. Do not place concrete until samples and product data for bar supports have been accepted by Engineer.

PART 2 PRODUCTS

2.01 DESIGN AND PERFORMANCE CRITERIA

- A. The drawings contain notes describing the size and spacing of reinforcement and its placement, details of reinforcement at wall corners and intersections, and details of extra reinforcement around openings in concrete, and other related information.

2.02 MATERIALS

- A. Reinforcing bars:
 - 1. Provide reinforcement of the grades and quality specified, fabricated from new stock, free from excessive rust or scale, and free from unintended bends or other defects affecting its usefulness.
 - 2. Reinforcing bars:
 - a. ASTM A615 Grade 60 deformed bars, including the following requirements, or ASTM A706 Grade 60 deformed bars.
 - 1) Actual yield strength based on mil tests of reinforcement provided shall not exceed the minimum yield strength specified in this Section by more than 18,000 pounds per square inch.
 - 2) Ratio of actual ultimate tensile strength to actual tensile yield strength shall not be less than 1.25.
 - 3. Reinforcing bars designated or required to be welded:
 - a. Low-alloy, ASTM A706 Grade 60, deformed bars.
- B. Bar supports:
 - 1. Wire supports:
 - a. All stainless steel bar supports:
 - 1) Conforming to CRSI Manual of Standard Practice recommendations for types and details, but custom fabricated entirely from stainless steel wire conforming to ASTM A493, AISI Type 316.
 - b. Stainless steel protected bar supports:
 - 1) Conforming to CRSI Manual of Standard Practice Class 2, Type B, and consisting of bright basic wire support fabricated from cold--drawn carbon steel wire with stainless steel ends attached at the bottom of each leg.
 - 2) Stainless steel wire ends shall conform to ASTM A493, AISI Type 316 and shall extend at least 3/4 inch inward from the formed surface of the concrete.
 - c. Bright basic wire bar supports.
 - 1) Conforming to CRSI Manual if Standard Practice, Class 3.
 - 2. Plastic supports:
 - a. Manufacturers: The following or equal:
 - 1) Aztec Concrete Accessories.
 - 3. Deformed steel reinforcing bar supports:
 - a. Fabricated of materials and to CRSI details recommended for typical reinforcement embedded in concrete and bent to dimensions required to provide specified clearances and concrete cover.
 - 4. Precast concrete bar supports ("dobies"):
 - a. Pre-manufactured, precast concrete blocks with cast-in annealed steel wires, 16-gauge or heavier.

- b. Compression strength of concrete: Equal to or exceeding the compression strength of the surrounding concrete.
 - c. Block dimensions:
 - 1) Height to provide specified concrete cover.
 - 2) Footprint not less than 3 inches by 3 inches, and adequate to support the weight of the reinforcement and maintain specified concrete cover without settling into the underlying surface.
 - 5. Stainless steel wire supports on stainless steel plates:
 - a. Type 304 stainless steel wire bar support chairs or bolsters supported on Type 304 stainless steel plates resting on the ground surface.
 - 1) Weld plates to at least 2 legs of wire support chairs.
- C. Tie wires:
 - 1. General use: Black annealed steel wire, 16-gauge or heavier.
- D. Welded wire fabric reinforcement:
 - 1. Material:
 - a. Carbon steel conforming to ASTM A1064.
 - 2. Provide welded wire reinforcement in flat sheet form. Rolled wire fabric is not permitted.
 - 3. Fabric may be used in place of reinforcing bars if accepted by Engineer:
 - a. Provide welded wire fabric having cross-sectional area per linear foot not less than the cross-sectional area per linear foot of reinforcing bars indicated on the Drawings.
- E. Mechanical reinforcing bar couplers:
 - 1. General:
 - a. Only products conforming to the requirements of ACI 318 for mechanical splices, and holding a current Evaluation Report that documents the following performance characteristics, will be considered for use.
 - b. Strength of coupler: Capable of developing tension and compression strength not lower than the lesser of the following:
 - 1) ACI 318 "Type 2" units: In static tension and compression:
 - a) Minimum 125 percent of the ASTM-specified minimum yield strength of the reinforcement being spliced.
 - b) Minimum 100 percent of the ASTM-specified minimum ultimate strength of the reinforcement being spliced.
 - c. Slip of reinforcing bars within coupler: Total slip of the reinforcing bars within the splice sleeve limited as follows:
 - 1) For bar sizes #14 and smaller, elongation between gauge points measured clear of the splice sleeve not exceeding 0.010 inches after coupler has been loaded to a tension of 30,000 pounds per square inch and load relaxed to a tension of 3,000 pounds per square inch.
 - d. Fabrication:
 - 1) Threaded joints:
 - a) Provide threaded ends designed so that cross-threading of bars will not occur during assembly.
 - b) Fabricate male ends for female couplers using coupler manufacturer's bar threading equipment to ensure proper taper and thread engagement.
 - 2) Mark each sleeve with heat treatment lot number.

2. Couplers: Threaded - Reinforcing bar splice at construction joints.
 - a. Steel sleeve butt splice with tapered internal threads in forged or swaged head, and nailing flange for attaching to forms. Provide with matching, tapered male-threaded dowels for insertion and tightening into threaded sleeve after form removal.
 - 1) Provide sleeve with factory-installed plugs to prevent concrete mortar from entering internally threaded coupler.
 - 2) Provide optional clipped nailing flanges as required to maintain minimum specified concrete cover over surfaces of coupler.
 - b. Holding current Evaluation Report demonstrating acceptance under ICC-ES AC133.
 - c. Manufacturers: One of the following or equal:
 - 1) Dayton Superior, DBDI Splice System.
 - 2) ERICO-Pentair, Lenton Form Saver.
3. Couplers: Threaded - reinforcing bar splice:
 - a. Steel sleeve butt splice with tapered internal threads at each end for joining to matching tapered male threads on reinforcing bars.
 - b. Holding current Evaluation Report demonstrating acceptance under ICC-ES AC133.
 - c. Manufacturers: One of the following, or equal:
 - 1) Dayton Superior: Taper-Lock System.
 - 2) ERICO-Pentair: Lenton Taper Threaded Splicing System.

2.03 FABRICATION

- A. Shop fabrication and assembly:
 1. Cut and bend bars in accordance with provisions of ACI 318 and the CRSI Manual of Standard Practice.
 2. Bend bars cold. Use bending collars to develop the recommended bend radius.
 3. Provide bars free from defects and kinks and from bends not indicated on the Drawings.
 4. Circumferential and radiused reinforcement: Roll to the radius required for its location in the structure before installation.
 5. Bars to be fitted with mechanical couplers:
 - a. Fabricate threaded ends for connections in shop using manufacturer's recommended tools. Field fabrication is not allowed.
 - b. Cut ends square.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verification of conditions:
 1. Reinforcing bars and welded wire reinforcement:
 - a. Verify that reinforcement is new stock, free from rust scale, loose mill scale, excessive rust, dirt, oil, and other coatings that will adversely affect bonding capacity when placed in the Work.
 2. Welded wire fabric:
 - a. Verify that sheets are not curled or kinked before or after installation.

3.02 PREPARATION

A. Surface preparation:

1. Reinforcing bars - uncoated:
 - a. Clean reinforcement of concrete, dirt, oil and other coatings that will adversely affect bond before embedding bars in subsequent concrete placements.
 - b. Thin coating of red rust resulting from short exposure will not be considered objectionable. Thoroughly clean bars having rust scale, loose mill scale, or thick rust coat.
 - c. Partially embedded reinforcement: Remove concrete or other deleterious coatings from dowels and other projecting bars by wire brushing or sandblasting before bars are embedded in subsequent concrete placements.

3.03 INSTALLATION

A. Reinforcing bars: General:

1. Field-cutting of reinforcing bars is not permitted.
2. Field-bending of reinforcing bars, including straightening and rebending, is not permitted.

B. Placing reinforcing bars:

1. Accurately place bars to meet position and cover requirements indicated on the Drawings and specified. Secure bars in position.
2. Tolerances for placement and minimum concrete cover: As listed in Table 1.

Table 1 - Reinforcement Placing Tolerances		
Member	Tolerance on Reinforcement Location ⁽¹⁾	Tolerance on Minimum Concrete Cover ^(1,2)
Slabs, beams, walls and columns except as noted below:		
10 inches thick and less	$\pm 3/8$ inch	- 3/8 inch
More than 10 inches thick	$\pm 1/2$ inch	- 1/2 inch
Formed soffits:	As noted above	- 1/4 inch
Longitudinal location of bends and ends of reinforcement:		
Conditions not listed below:	± 2 inches	- 1/2 inch
At discontinuous ends of brackets and corbels	$\pm 1/2$ inch	- 1/4 inch
At discontinuous ends of other members:	± 1 inch	- 1/2 inch
Notes:		
(1) $\pm$ indicates "plus or minus;" - indicates "minus;" + indicates "plus."		
(2) Tolerance on cover is limited as noted, but decrease in cover shall not exceed one third of the minimum cover indicated on the Drawings.		

3. Spacing between bars:
 - a. Minimum clear spacing between bars in a layer:
 - 1) As indicated on the Drawings, but not less than the larger of 1.5 times the bar diameter or 1-1/2 inches.
 - b. Minimum clear spacing between bars in 2 or more parallel layers:
 - 1) Place bars in upper layers directly above bars in lower layers.
 - 2) Minimum spacing between layers: As indicated on the Drawings, but not less than the larger of 1.5 times the bar diameter or 1-1/2 inches.
 - c. Limits on minimum clear spacing between bars also applies to the clear spacing between a lap splice and the adjacent bars and/or lap splices.
 4. Lap splices for bars:
 - a. Lap splice locations and lap splice lengths: as indicated on the Drawings. Where lap lengths are not indicated, provide in accordance with ACI 318.
 - b. Unless otherwise specifically indicated on the Drawings (and noted as "non-contact lap splice"), install bars at lap splices in contact with each other and fasten together with tie wire.
 - c. Where bars are to be lap spliced at concrete joints, ensure that bars project from the first concrete placement a length equal to or greater than minimum lap splice length indicated on the Drawings.
 - d. Stagger lap splices where indicated on the Drawings.
 - e. Where lap splice lengths are not indicated on the Drawings, provide lap splice lengths in accordance with ACI 318.
- C. Reinforcing supports:
1. Provide supports of sufficient numbers, sizes, and locations to maintain concrete cover, to prevent sagging and shifting, and to support loads during construction without displacement and without gouging or indentation into forming surfaces.
 - a. Quantities and locations of supports shall not be less than those indicated in ACI SP-66 and the CRSI Manual of Standard Practice.
 2. Do not use brick, concrete masonry units, concrete spalls, rocks, wood, or similar materials for supporting reinforcement.
 3. Do not use "give away bars" that have less cover than that required by the Contract Documents. Do not adjust the location of reinforcement required by the Contract Documents to provide cover for give away bars.
 4. Provide bar supports of height required to maintain the clear concrete cover indicated on the Drawings.
 5. Provide bar supports at formed vertical faces to maintain the clear concrete cover indicated on the Drawings.

6. Schedule of reinforcement support materials: Provide bar supports as indicated in Table 2.

Table 2 - Reinforcement Support Materials		
Case	Location	Material
a.	Concrete placed over earth and concrete seal slabs ("mud mats"):	Precast concrete bar supports.
b.	Concrete placed against forms and exposed to water or wastewater process liquids (whether or not such concrete received additional linings or coatings):	All stainless steel bar supports.
c.	Concrete placed against forms and exposed to earth, weather, frequent washdown, or groundwater in the finished work:	All stainless steel bar supports.
d.	Concrete placed against forms and exposed to interior equipment/piping areas in the finished work:	All stainless steel bar supports.
e.	Between mats of reinforcement, and fully embedded within a concrete member:	Bright basic wire bars supports, or deformed steel reinforcing bars.

- D. Tying of reinforcing:
1. Fasten reinforcement securely in place with wire ties.
 2. Tie reinforcement at spacings sufficient to prevent shifting.
 - a. Provide at least 3 ties in each bar length. (Does not apply to dowel lap splices or to bars shorter than 4 feet, unless necessary for rigidity).
 3. Tie slab bars at every intersection around perimeter of slab.
 4. Tie wall bars and slab bar intersections other than around perimeter at not less than every fourth intersection, but at not more than the spacing indicated in Table 3:

Table 3 - Maximum Spacing of Tie Wires for Reinforcement		
Bar Size	Slab Bar Spacing (inches)	Wall Bar Spacing (inches)
Bars Number 5 and Smaller	60	48
Bars Number 6 through Number 9	96	60
Bars Number 10 and Number 11	120	96

5. After tying:
- a. Bend ends of wires inward towards the center of the concrete section. Minimum concrete cover for tie wires shall be the same as cover requirements for reinforcement.
 - b. Remove tie wire clippings from inside forms before placing concrete.

- E. Welded wire fabric reinforcement:
1. Install only where indicated on the Drawings or accepted in advance by Engineer.
 2. Install necessary tie wires, spacing chairs, and supports to keep welded wire fabric at its designated position in the concrete section while concrete is being placed.
 3. Straighten welded wire fabric to make sheets flat in the Work.
 4. Do not allow wire fabric to drape between supports unless such a configuration is specifically indicated on the Drawings.
 - a. If fabric is displaced during placement of concrete, make provisions to restore it to the designated location using methods acceptable to Engineer.
 5. Bend welded wire fabric as indicated on the Drawings or required to fit Work.
 6. Lap splice welded wire fabric as indicated on the Drawings.
 - a. If lap splice length is not indicated, splice in accordance with ACI 318, but not less than 1 1/2 courses of fabric or 8 inches minimum. Tie laps at ends and at not more than 12 inches on center.
- F. Welding reinforcing bars:
1. Weld reinforcing bars only where indicated on the Drawings or where acceptance is received from Engineer prior to welding.
 2. Perform welding in accordance with AWS D1.4 and welding procedures accepted by Engineer.
 - a. Conform to requirements for minimum preheat and interpass temperatures.
 3. Submit:
 - a. Welding procedures specification.
 - b. Procedures qualification record.
 - c. Welder qualification test record.
 4. Do not tack weld reinforcing bars except where specifically indicated on the Drawings.
- G. Reinforcing bar mechanical couplers:
1. Install only at locations indicated on the Drawings or where prior approval has been obtained from Engineer.
 2. Install in accordance with manufacturer's instructions and requirements of Evaluation Report.
 - a. Make splices using manufacturer's standard equipment, jigs, clamps, and other required accessories.
 - b. After assembly of the splice, tighten using torque load not less than that recommended by the manufacturer.
 3. Unless greater cover is indicated on the Drawings, provide clear cover from surface of concrete to outside face of couplers that is not less than the minimum concrete cover specified for typical reinforcement.
 - a. If cover is less than required, contact Engineer for evaluation of conditions before modifying locations of bars or placing concrete.
 - b. Modifications to maintain or provide required concrete cover, such as addition of concrete ; re-positioning of stirrups, ties, etc., may be completed only after approval by Engineer.

3.04 FIELD QUALITY CONTROL

- A. Provide quality control for the Work of this Section as specified in Section 01450 -Quality Control.
- B. Field inspections and testing:
 - 1. Submit records of inspections and testing to Engineer in electronic format within 24 hours after completion.
- C. Manufacturer's services:
 - 1. Furnish manufacturer's technical representative to conduct jobsite training regarding proper storage, handling, and installation of mechanical reinforcing bar couplers for personnel who will perform the installation. Engineer may attend training session.

3.05 FIELD QUALITY ASSURANCE

- A. Provide quality assurance as specified in Section 01450 - Quality Control.
- B. Special inspections and tests:
 - 1. Provide as specified in Section 01455 - Regulatory Quality Assurance.
 - 2. Frequency of inspections:
 - a. Unless otherwise indicated on the Drawings or in this Section, provide periodic special inspection as required by the Building Code specified in Section 01410 - Regulatory Requirements.
 - 3. Preparation:
 - a. Review Drawings and Specification for the Work to be observed.
 - b. Review approved submittal and shop drawings.
 - 4. Inspections: Special inspection shall include, but is not limited to, the following items.
 - a. Reinforcement: General:
 - 1) Type (material) and location of reinforcement supports.
 - 2) Bar material/steel grade and bar size.
 - 3) Location, placement, and spacing of bars.
 - 4) Clear concrete cover over reinforcement.
 - 5) Lap splice: Location and lap length. Bars within tolerances for contact (unless non-contact splice is indicated on the Drawings).
 - 6) Bar hooks and development lengths embedded within concrete sections as indicated on the Drawings.
 - 7) Reinforcement tied in position and tie wire legs turned inward toward the center of the concrete section.
 - b. Reinforcement: Welding:
 - 1) Inspector qualification and inspections shall be in accordance with the requirements of AWS D1.4.
 - 2) Provide periodic inspection for:
 - a) Weldability of reinforcement other than ASTM A706.
 - b) Single pass fillet welds with thickness less than or equal to 5/16 inch.
 - 3) Provide continuous inspection for:
 - a) Other welds.
 - b) Welds at mechanical reinforcing bar couplers and end anchors.

- 4) In addition to visual inspection, Owner may inspect reinforcing bar welds by other methods, including radiographic inspection.
- 5. Mechanical reinforcing bar couplers:
 - a. Special inspection shall include, but is not limited to, the following items:
 - 1) Coupler model and identification.
 - 2) Couplers are installed in accordance with the requirements of the Engineering Report for each product.
 - 3) Confirmation of the following:
 - a) Grade and size of reinforcing bars.
 - b) Position of couplers.
 - c) Insertion length of reinforcement.
 - d) Tightening of bars in the couplers.
- 6. Records of inspections:
 - a. Provide a written record of each inspection using forms acceptable to the Engineer and to the Building Official.
 - b. Submit electronic copies of inspection reports to Engineer within 24 hours after completion of inspections.

3.06 NON-CONFORMING WORK

- A. Before placing concrete, adjust or remove and re-install reinforcement to conform to the requirements of the Contract Documents.

END OF SECTION

SHOP DRAWING

Nan, Inc

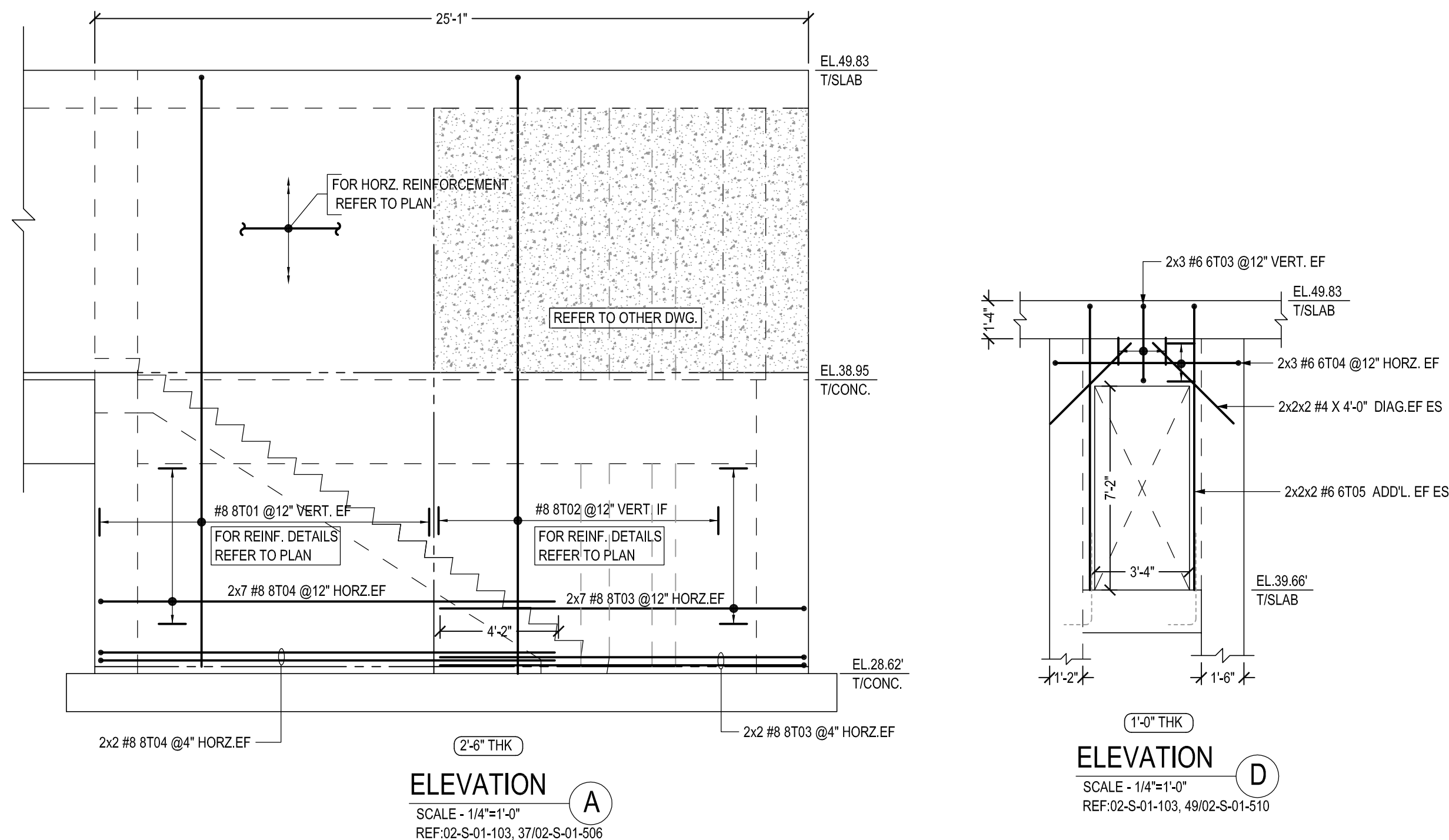
PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

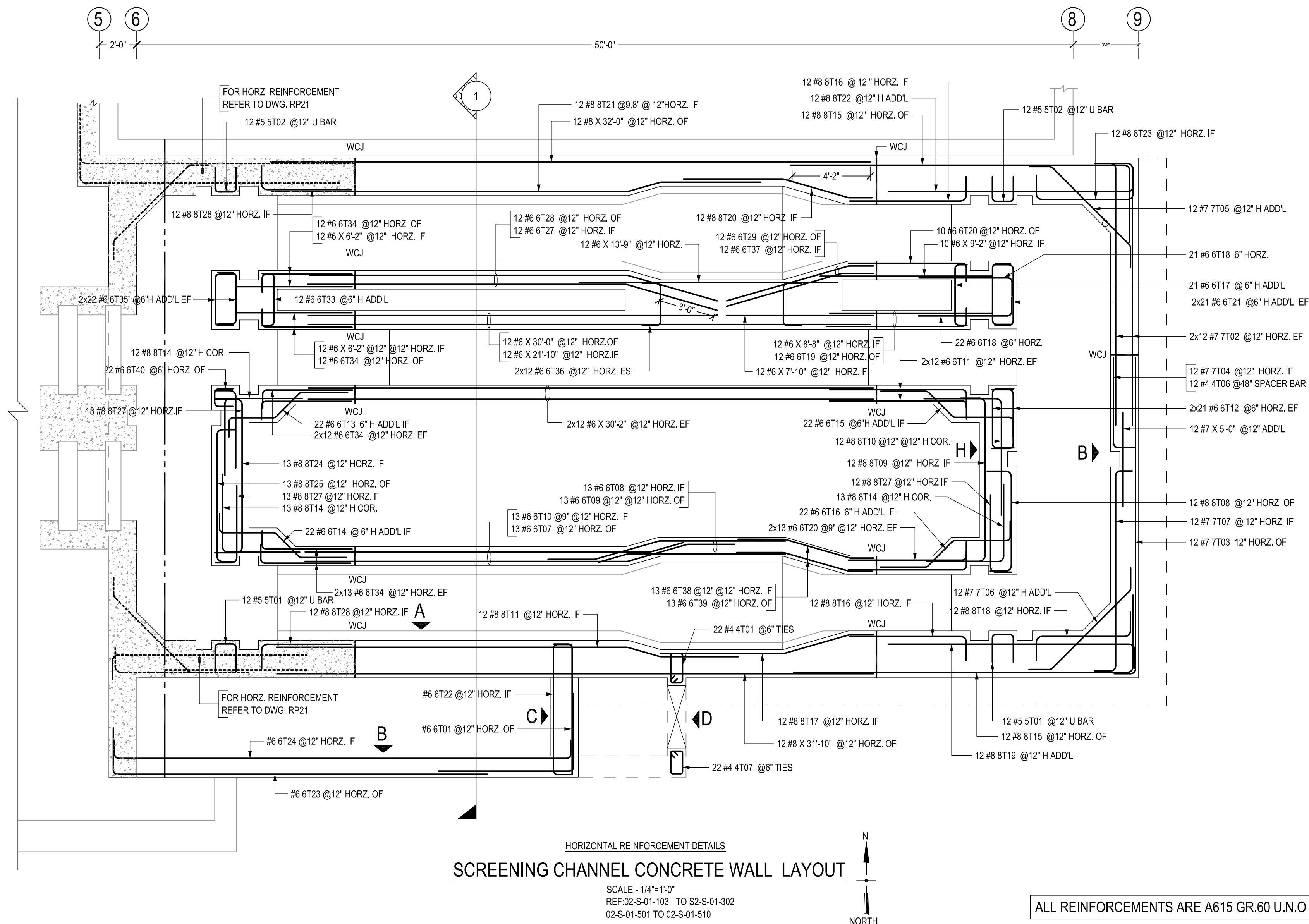
THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/17/2025
SPECIFICATION SECTION # 03200
SPECIFICATION CONCRETE REINFORCING
PARAGRAPH 1.04
DRAWING SCREENING CHANNEL
SUBCONTRACTOR _____
SUPPLIER CMC REBAR
MANUFACTURER _____

CERTIFIED BY: *Nathan*



1'-0" THK
ELEVATION
SCALE - 1/4"=1'-0"
REF:02-S-01-103, 49/02-S-01-510



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SUBMITTAL RECORD

DATE: 09/10/25

FOR: APPROVAL

DATE: .

FOR: .

DATE: .

FOR: .

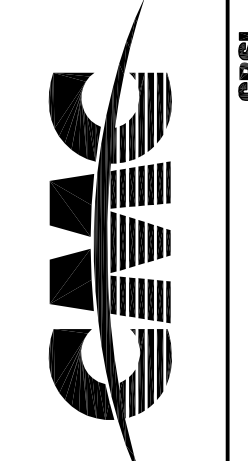
Δ DATE: .

Δ DATE: .

JOB #	2524240001
DETAILER:	MKR
EMAIL:	will@memasterrebor.com
PHONE:	(360)-346-6729
REF:	

PROJECT INFORMATION
ALL REINFORCING

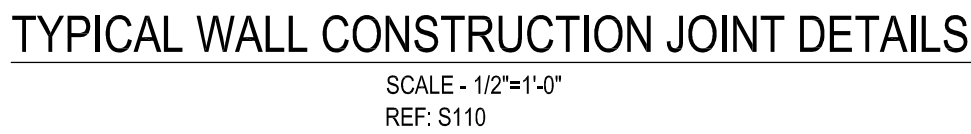
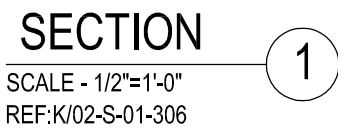
CUSTOMER:	
JOB NAME:	HILO WWTP
LOCATION:	COUNTY OF HAWAII
DESCRIPTION:	CORRECTION

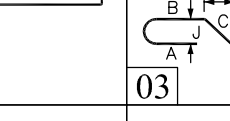
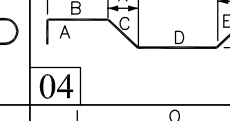
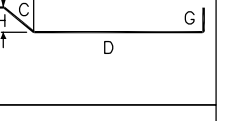
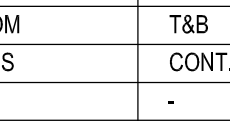
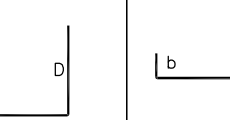
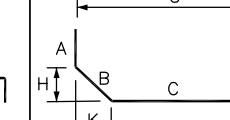
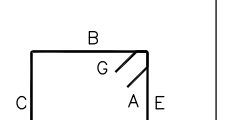
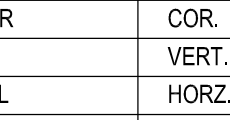
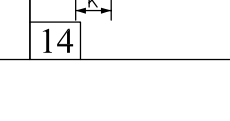
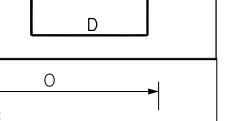
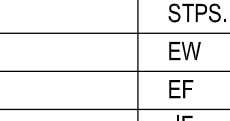
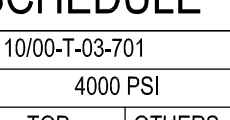
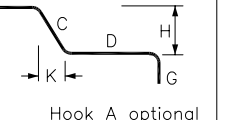
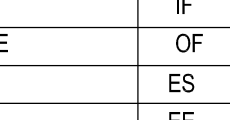
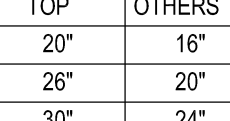
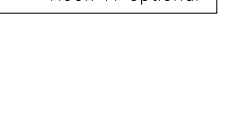
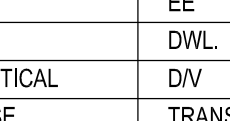
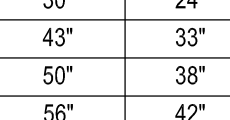
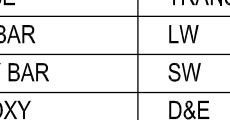


JOB# 2524240001

DRAWING:

RP-033



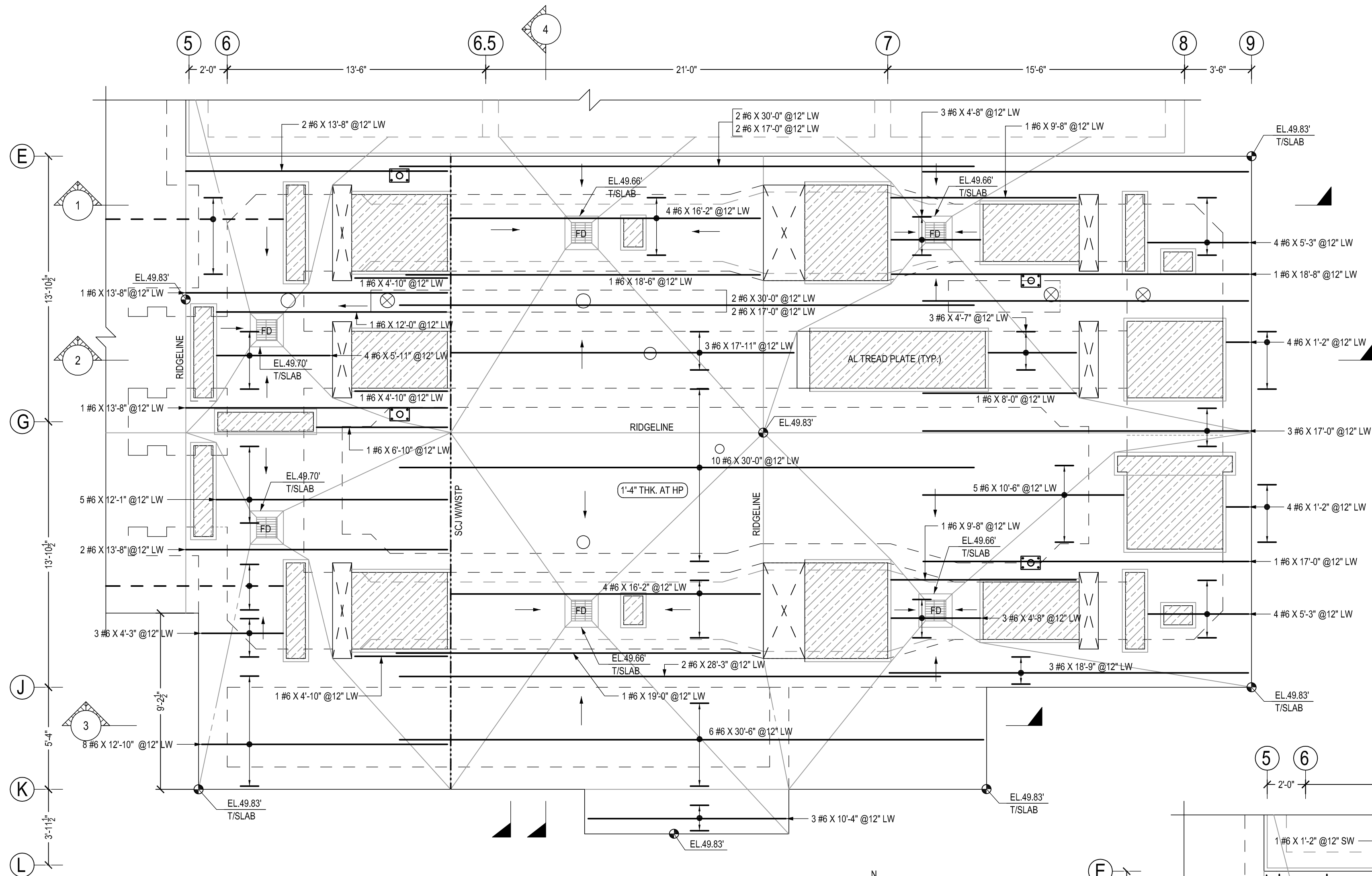
																																																																																																															
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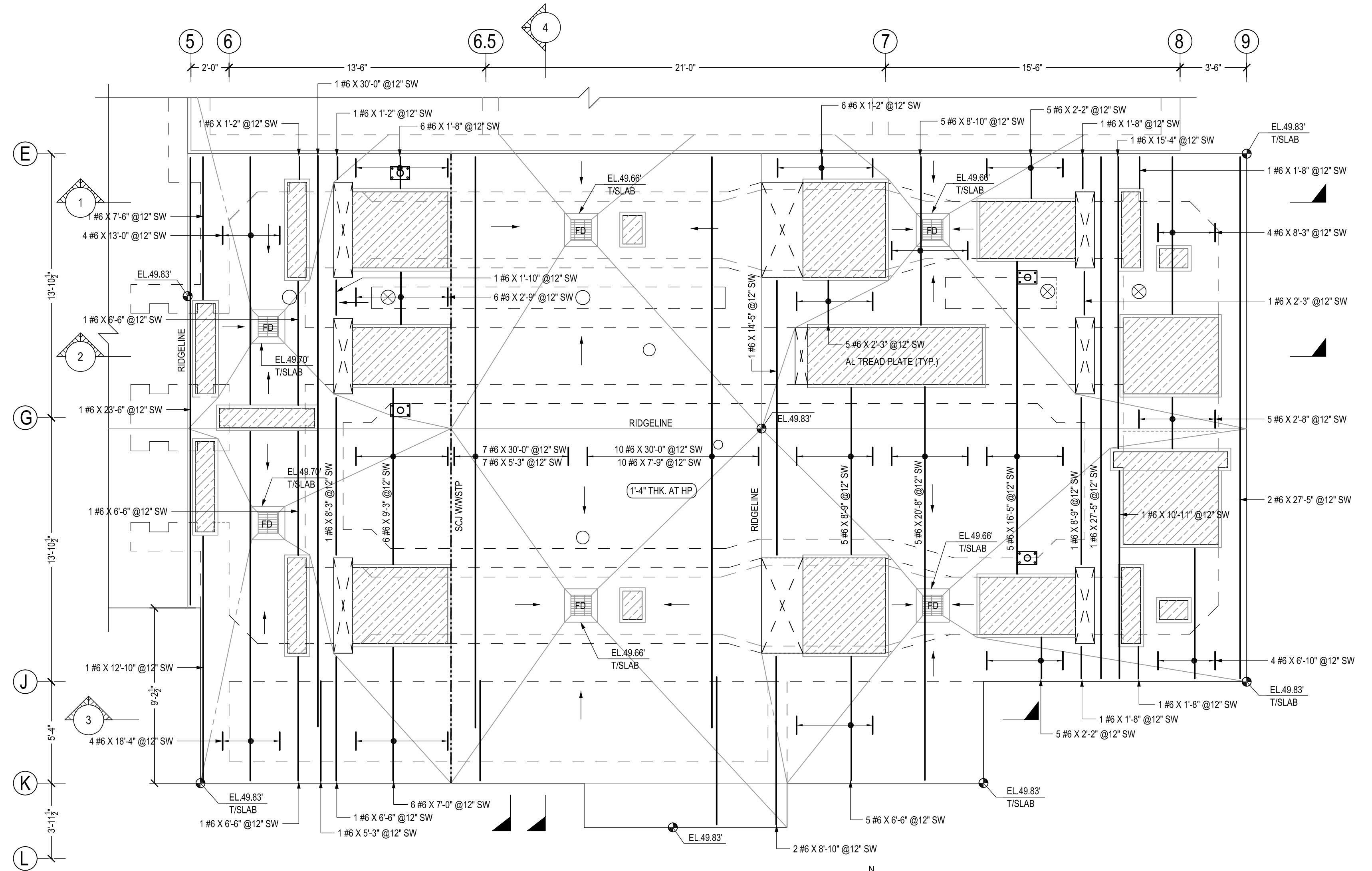
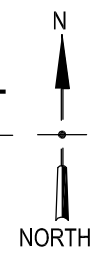
FIELD USE: THIS DRAWING TO BE USED IN CONJUNCTION W/ STRUCTURAL AND ARCHITECTURAL DOCUMENTS TO FACILITATE PLACING OF REINFORCING BARS ONLY. DO NOT USE THIS DRAWING FOR LAYOUT OR CONSTRUCTION. ANY DIMENSIONS OR ELEVATIONS NOTED IN THIS DOCUMENT ARE FOR REFERENCE ONLY. TO FACILITATE APPROVAL &/OR PLACEMENT OF REBAR, DISCREPANCIES SHOULD BE NOTED AT THE TIME OF APPROVAL. CMC REBAR ASSUMES NO LIABILITY FOR DIMENSIONAL ERRORS IN THE LAYOUT OF THE BUILDING OR FORMS.

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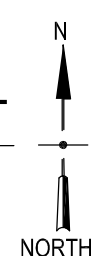
[illegible]



TOP LONG WAY REINFORCEMENT DETAILS
SCREENING TOP SLAB REINFORCEMENT LAYOUT
SCALE - 1/4"=1'-0"
REF:02-S-01-104



TOP SHORT WAY REINFORCEMENT DETAILS
SCREENING TOP SLAB REINFORCEMENT LAYOUT
SCALE - 1/4"=1'-0"
REF:02-S-01-104



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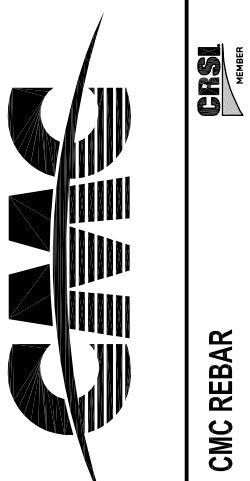
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SUBMITTAL RECORD

DATE	09/10/25	FOR	APPROVAL
DATE		FOR	
DATE		FOR	
DATE		FOR	
DATE		FOR	
DATE		FOR	

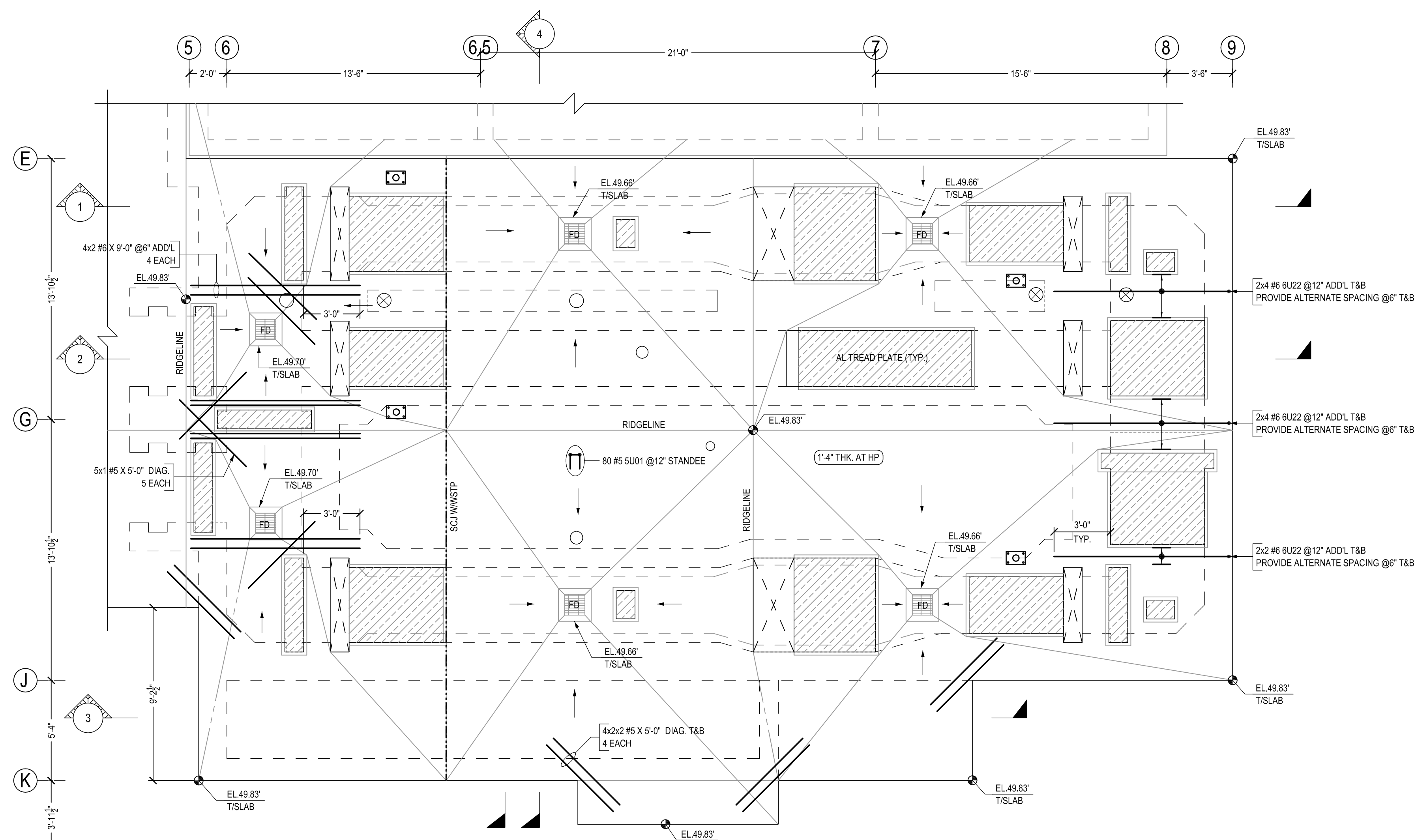
PROJECT INFORMATION

CUSTOMER	2524240001	FOR	APPROVAL
JOB NAME	HILO WWTP	FOR	
LOCATION	COUNTRY OF HAWAII	FOR	
DESCRIPTION	SCREENING - TOP SLAB REINFORCING	FOR	
DETAILS	will@mcsherrill.com	FOR	
PHONE	(360)-346-6729	FOR	
DATE		FOR	
DATE		FOR	

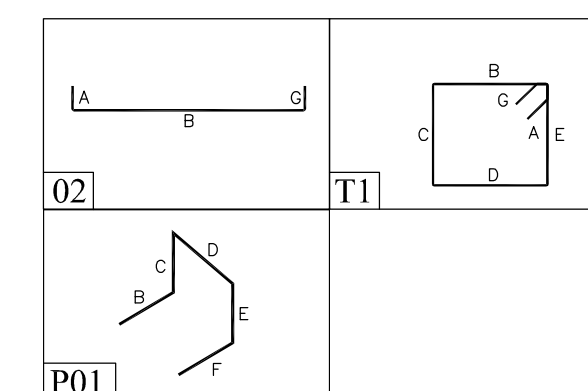


JOB# 2524240001

DRAWING: RP-036



Release Code: RP-036_RP-038						BENDING DETAILS.									
Bar Mark	Qty	Size	Total length type	A'	B'	C'	D'	E'	F'	G'	H'	J'	K'	L'	M'
4U01	40	#4	4'11"	11'	4 ½"	10"		11"	10"						
4U02	9	#4	5'11"	11'	4 ½"	10"	12"	10"	12"		4 ½"	3"			
5U01	80	#5	5'6"	P01		16"		11"	8"		16"				
6U01	16	#6	2'2"	2	10"		12"								
6U02	1	#6	8'6"	2	10"		7'6"								
6U03	4	#6	14'9"	2	10"		13'0"								
6U04	10	#6	2'8"	2	10"		1'8"								
6U05	7	#6	7'6"	2	10"		6'6"								
6U06	9	#6	13'10"	2	10"		12'10"								
6U07	4	#6	19'4"	2	10"		18'4"								
6U08	8	#6	6'9"	2	10"		5'9"								
6U09	10	#6	30'0"	2	10"		29'0"								
6U10	1	#6	25'6"	2	10"		23'6"					10"			
6U11	3	#6	29'5"	2	10"		27'5"					10"			
6U12	1	#6	16'4"	2	10"		15'4"								
6U13	14	#6	9'3"	2	10"		8'3"								
6U14	1	#6	11'11"	2	10"		10'11"								
6U15	4	#6	7'10"	2	10"		6'10"								
6U16	10	#6	3'2"	2	10"		2'2"								
6U17	5	#6	21'8"	2	10"		20'8"								
6U18	7	#6	9'10"	2	10"		8'10"								
6U19	8	#6	6'3"	2	10"		5'3"								
6U20	8	#6	17'0"	2	10"		16'0"								
6U21	4	#6	19'8"	2	10"		18'8"								
6U22	20	#6	10'4"	2	10"		9'4"								
6U23	6	#6	31'0"	2	10"		30'0"								
6U24	6	#6	8'0"	2	10"		7'0"								
6U25	3	#6	5'3"	2	10"		4'3"								
6U26	3	#6	12'4"	2	10"		10'4"				10"				



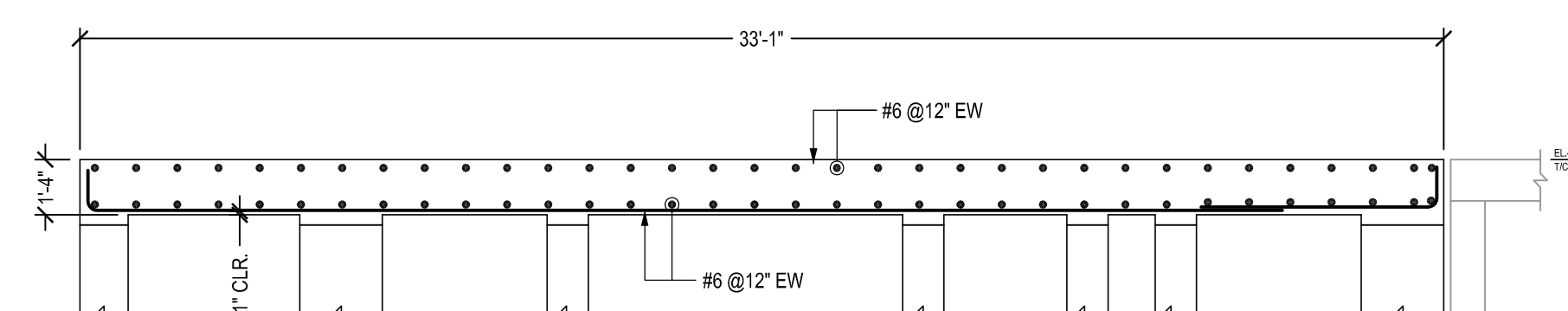
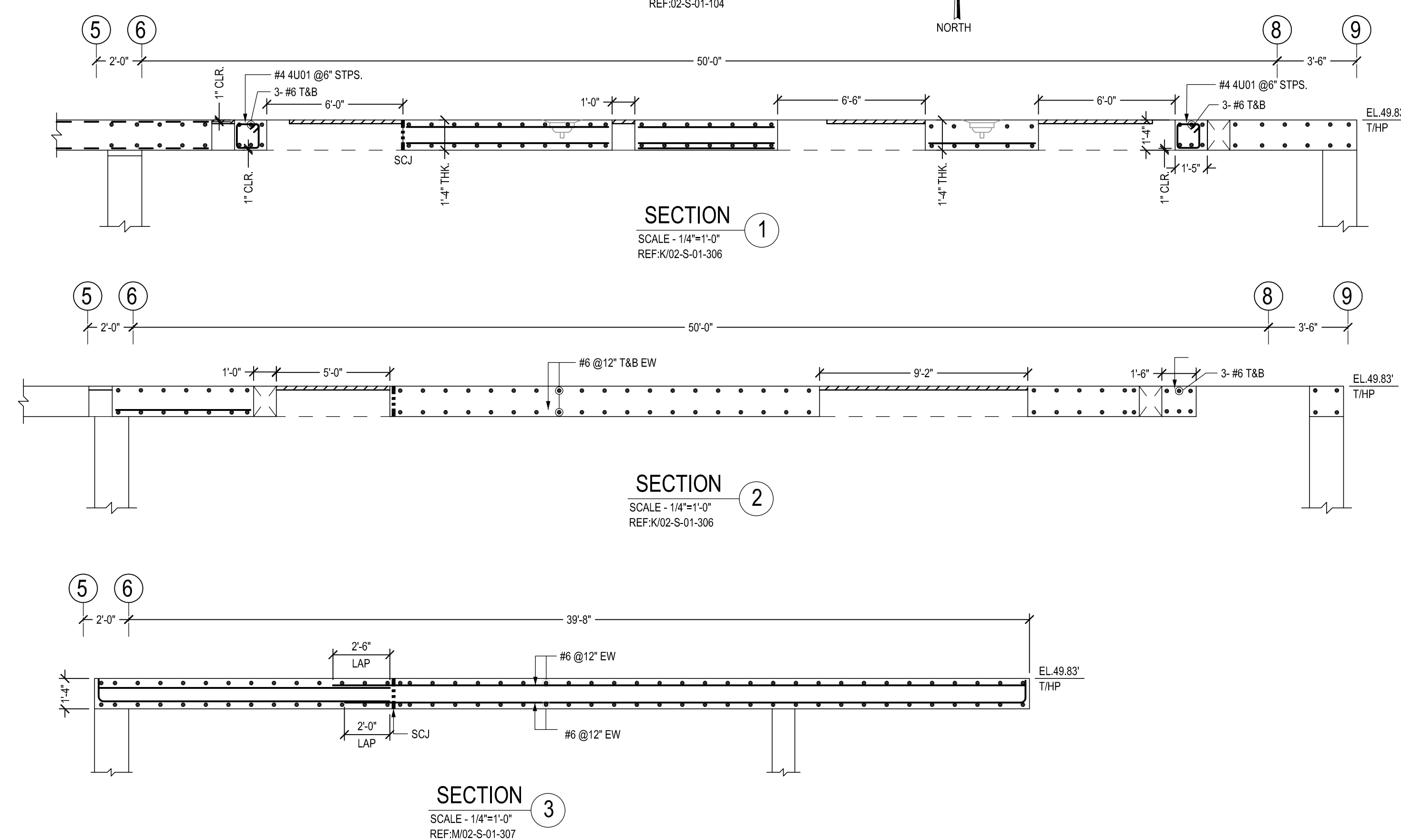
ABBREVIATIONS	
BOTTOM	BOT.
TOP&BOTTOM	T&B
CONTINUOUS	CONT.
EACH WAY	EW
EACH FACE	EF
INNER FACE	IF
OUTER FACE	OF
EACH SIDE	ES
DOWEL	DWL.
DOWEL/VERTICAL	D/V
LONG WAY BAR	LW
SHORT WAY BAR	SW
DRILL & EPOXY	D&E
REFERENCE	REF.

LAP SCHEDULE		
REF: S10100-T-03-701		
SIZE	4000 PSI	
	TOP	OTHERS
#4	20"	16"
#5	26"	20"
#6	30"	24"
#7	43"	33"
#8	50"	38"
#9	56"	42"
#10	62"	48"
#11	69"	54"

ADDITIONAL REINFORCEMENT DETAILS

SCREENING TOP SLAB REINFORCEMENT LAYOUT

SCALE - 1/4"=1'-0"
REF:02-S-01-104



SECTION 4

SCALE - 1/4"=1'-0"
REF:M/02-S-01-307

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